

BIOINFORMATICS INDUSTRIAL RESEARCH TRAINING PROGRAM

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DEGREE OF BACHELOR OF SCIENCE
IN
BIOTECHNOLOGY

SUBMITTED BY
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DEPARTMENT OF BIOTECHNOLOGY
JAMSHEDPUR WOMEN'S UNIVERSITY, JAMSHEDPUR, JHARKHAND
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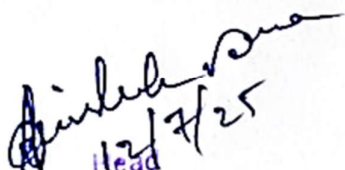
Date: 12.07.2025

To Whomsoever It May Concern

This is to certify that **Miss Afreen Parween** has undergone 90 days Industrial Professional Advanced Bioinformatics & Genomics Internship Program 2025 at Dept. of Tech Biosciences, Digianalix where she successfully completed training program in Building Concepts of Bioinformatics, working in Genomics, DNA microarray analysis, system biology, Python, nutrition and health with herbal plants. Computational Drug design and docking.

She worked on research project entitled **"Integrative molecular docking approaches in targeting genomic alterations from whole genome & exome sequencing in pancreatic cancer"** in bioinformatics facilities at Dept. of Tech. Biosciences, Digianalix Tech. Institute & Research Solutions. She had also contributed in a company project on nutrition and health added as review article entitled **"Medicinal plants and herbs which plays an important role in aging and osteoporosis"** and **"Metabolomics approaches of identifying potential metabolites present in effective medicinal plants related to aging"** is under communicated for publication. She has successfully completed Internship & Research Training Program with an excellent performance.

During her internship tenure she learned good communication skills, collaboration and drafting research article. She preferred to take challenges in research projects along with good collaborative work nature. We wish her all the best in her future career.


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CERTIFICATE FROM DEPARTMENT

This is to certify that the dissertation project entitled “**INTEGRATIVE MOLECULAR DOCKING APPROACHES IN TARGETING GENOMIC ALTERATIONS FROM WHOLE GENOME & EXOME SEQUENCING IN PANCREATIC CANCER**” was submitted by **AFREEN PARWEEN** a student of B.Sc. Biotechnology, bearing the registration number JWU2022041703 of JAMSHEDPUR WOMEN'S UNIVERSITY, JAMSHEDPUR and exam roll number 223204913176 has successfully completed the work for partial fulfilment of Graduation. This work is carried out under the supervision of H.O.D. Dr. Vishwa Raj Lal (Assistant Profession) of JWU, Jamshedpur and Dr. Mukesh Nitin, Scientist & Head, Department of Tech. Bioscience & Data Science at Digianalix (Tech. Institute and Research Solutions). It is further testified that no part of this work has been submitted for any other degree or Diploma.

EXAMINER

INTERNAL SUPERVISER

EXTERNAL SUPERVISOR

DECLARATION

I AFREEN PARWEEN hereby certify that the internship report entitled **“BIOINFORMATICS INDUSTRIAL INTERNSHIP & RESEARCH TRAINING- PROGRAM”** prepared by me under the guidance of Dr Mukesh Nitin, Scientist & Head, Department of Tech.- Bioscience & Data Science at Digianalix (Tech. Institute and Research Solutions).

I also declare that this Internship work is towards the partial fulfillment of the university regulations for the award of a Bachelor's of Science in Biotechnology by JAMSHEDPUR WOMEN'S UNIVERSITY, JAMSHEDPUR, JHARKHAND.

I have undergone a three-month Research and training program. I further declare that this research is based on my original study and has not been submitted for the award of any degree or diploma from any other institution.

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Roll no- 223204917

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I offer a sincere appreciation to **Dr Mukesh Nitin**, Visiting Scientist and project Coordinator of Tech. Biosciences & Data Sciences Department at Digianalix (Tech. Institute and Research Solutions), for his invaluable assistance and guidance in writing this project. His timely counsel and methodical approach have aided me in completing this assignment.

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SECTION A

Industrial Training

1. BIOINFORMATICS

1.1 INTRODUCTION TO BIOINFORMATICS

Bioinformatics is an interdisciplinary field mainly involving molecular biology and genetics, computer science, mathematics, and statistics. Data intensive, large-scale biological problems are addressed from a computational point of view. The most common problems are modeling biological processes at the molecular level and making inferences from collected data. It is an interdisciplinary subject/ field/ research area which involves the technology that uses computers for Storage, Retrieval, Manipulation and Distribution of information related to biological macromolecules such as DNA, RNA and proteins. Additional information includes the text of scientific papers and "relationship data" from metabolic pathways, taxonomy trees, and protein-protein interaction networks. Bioinformatics employs a wide range of computational techniques including sequence and structural alignment, database design and data mining, macromolecular geometry, phylogenetic tree construction, prediction of protein structure and function, gene finding, and expression data clustering. It involves the development of algorithms, databases, and software tools to understand biological processes at the molecular level through computational methods.

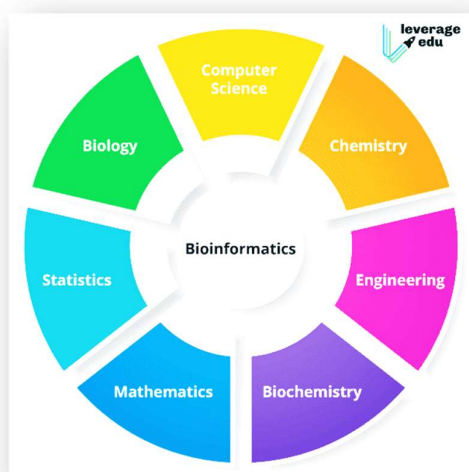


Fig.: **FIELDS INVOLVED IN BIOINFORMATICS**

(Image sources: <https://blogassets.leverageedu.com/blog/wp-content/uploads/2019/12/09001543/Bioinformatics-600x600.png>)

1.2 SCOPE OF BIOINFORAMATICS

The scope of Bioinformatics includes:

Healthcare Industries: Bioinformatics professionals contribute to medical diagnostics, personalized medicine, and clinical decision support systems in healthcare settings.

Pharmaceutical Companies: Bioinformatics plays an important role in drug discovery, target identification, and optimization of therapeutic compounds in pharmaceutical research and development.

Data Science: Bioinformatics provides a foundation for data science careers, with applications in analyzing large biological datasets, developing predictive models, and extracting actionable insights.

Research and Development: Bioinformatics professionals are involved in various research areas, including genomics, proteomics, systems biology, and structural biology, driving innovation and discovery.

Drug Designing and Development: Bioinformatics techniques are utilized in rational drug design, virtual screening of compound libraries, and optimizing drug efficacy and safety profiles.

Clinical Research: Bioinformatics supports clinical research endeavors, including biomarker discovery, disease modeling, and pharmacogenomics studies, contributing to improved patient outcomes and healthcare practices.

Software Development: Bioinformatics software developers create tools and algorithms for analyzing biological data, enhancing research capabilities and facilitating data-driven decision-making.

Computational Biology Bioinformatics professionals with expertise in computational biology apply mathematical and computational techniques to study biological systems, uncovering underlying principles and mechanisms.

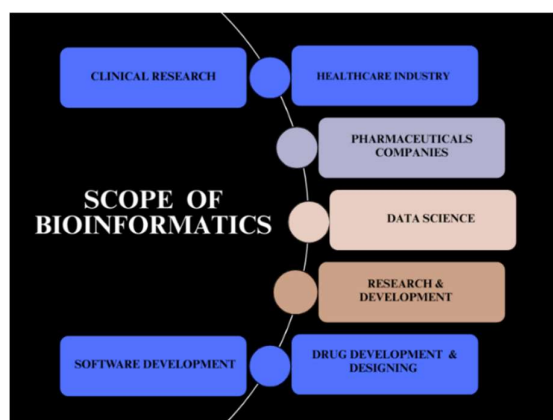


Fig: - **SCOPE OF BIOINFORMATICS**

1.3.APPLICATION OF BIOINFORMATICS

Some of the applications of Bioinformatics are:

- **Genomic Analysis**

Computational bioinformatics tools are employed to deal with the correspondence between the sequences of DNA and the genes, the regulatory genome, and the genetic variations that could exist in diseases or traits. Genome-wide association studies (GWAS) use bioinformatics approaches, which are the tools for determining genetic variation that may lead to complex trait formation.

- **Transcriptomics**

The transcriptomics research, which uses technologies such as RNA sequencing (RNA-seq), studies the patterns of gene expression. Bioinformatics tools allow RNA-seq information to be analyzed for different kinds of genes that involve differentially expressed genes, alternative splicing events, and non-coding RNAs, providing information about cells and processes.

- **Proteomics**

The field of proteomics is devoted to the analysis of proteins structure, function, and relationships within a biological system. The bioinformatics tools are being incorporated for multipurpose purposes such as the

identification, quantification, and structural prediction of proteins, the analysis of protein-protein interactions, and post-translational modifications (PTMs).

- **Metagenomics**

Metagenomics is about studying the communal genome of microbial communities as they stand in soil or bodies environment. Employing bioinformatics techniques, one can process metagenomic data as well as study various microbial species, predict metabolic pathways, and investigate microbial ecology and evolution.

- **Structural Bioinformatics**

Structural bioinformatics is exactly the branch of bioinformatics that studies not only modeling and analyzing but also predicting the biomolecular structure (for instance, proteins and nucleic acids). Bioinformatics approaches are used for the prediction of protein structure, molecular docking, and virtual screening of drugs as well.

- **Phylogenetics**

Phylogenetics, which entails depicting how organisms may have related through the generations by using data received by genetic studies, belongs to the study area of evolutionary biology. Several bioinformatics approaches, e.g., multiple sequence alignment and phylogenetic tree inference, are used to characterize DNA or protein sequences, yielding different information about evolutionary history and biodiversity.

- **Clinical Genomics**

Clinics and research institutions use bioinformatics to search the genetic data of their patients, determining whether they have a genetic disorder, recognizing the risk of a disease, and prescribing personalized treatments to them. This involves delineating mutations that are correlated with a disease, interpreting their significance, and suggesting personalized care



Fig.: - **APPLICATION OF BIOINFORMATICS**

2. DATABASE MANAGEMENT SYSTEM

2.1 INTRODUCTION TO DBMS

A **Database Management System (DBMS)** is a software solution designed to efficiently **manage, organize, and retrieve data** in a structured manner. It serves as a critical component in **modern computing, enabling** organizations to **store, manipulate,** and secure their data effectively. From small applications to enterprise systems, DBMS plays a vital role in supporting **data-driven decision-making** and operational efficiency. A DBMS is a system that allows users to **create, modify,** and query databases while ensuring **data integrity, security,** and efficient data access. Unlike **traditional file systems,** DBMS minimizes **data redundancy,** prevents inconsistencies, and simplifies data management with features like concurrent access and **backup mechanisms.** It organizes data into **tables, views, schemas,** and reports, providing a structured approach to data management.

2.2 Types of DBMS

There are several types of Database Management Systems (DBMS), each tailored to different data structures, scalability requirements, and application needs. The most common types are as follows:

I. Relational Database Management System (RDBMS)

RDBMS organizes data into tables (relations) composed of rows and columns. It uses primary keys to uniquely identify rows and foreign keys to establish relationships between tables. Queries are written in **SQL (Structured Query Language)**, which allows for efficient data manipulation and retrieval.

Examples: MySQL, Oracle, Microsoft SQL Server and PostgreSQL.

II. Object-Oriented DBMS (OODBMS)

OODBMS integrates object-oriented programming concepts into the **database environment**, allowing data to be stored as objects. This approach supports complex data types and relationships, making it ideal for applications requiring advanced data modeling and **real-world simulations**.

Examples: ObjectDB, db4o.

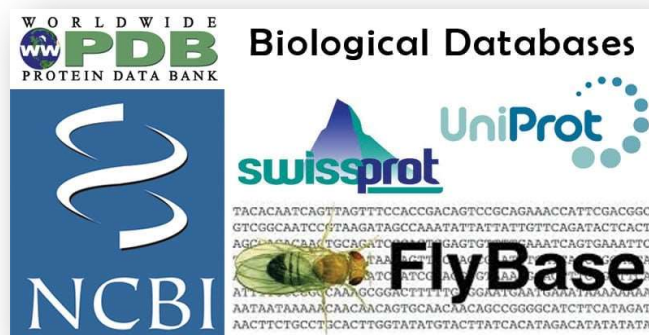
3. BIOLOGICAL DATABASES

3.1 Definition

These are the databases consisting of biological data like protein sequencing, molecular structure, DNA sequences, etc. in an organized form. Several computer's tools are there to manipulate the biological data like an update, delete, insert, etc. Scientists and researchers from all over the world enter their experiment data and results in a biological database so that it is available to a wider audience. Biological databases are free to use and contains a huge collection of variety of biological data. These data work closely with bioinformatics, which is a recent discipline focusing on addressing the need to analyze and interpret vast amounts of genomic data.

3.2 Some Databases sources used to collect information:

- i. **NCBI** National Centre for Biotechnology Information
- ii. **EMBL EBI** European Bioinformatics Institute.
- iii. **KEGG**-Kyoto Encyclopedia of Genes and Genome
- iv. **DDBJ** DNA Data Bank of Japan
- v. **PDB** Protein Data Bank
- vi. **PIR** Protein Information Resource
- vii. **OMIM** Online Mendelian Inheritance in Man
- viii. **SRS**-Sequence Retrieval System
- ix. **ENTREZ**-Entrez Global Query
- x. **SCOP**-Structural Classification of Proteins Databases
- xi. **GenBank**
- xii. **STRING**
- xiii. **PubMed**
- xiv. **UniProt**



*Fig: **DATABASES***

(Image sources: - <https://microbenotes.com/wp-content/uploads/2018/12/Biological-Databases-Types-and-Importance.jpg>)

3.3 Use of Biological Databases:

- i. Act as storage information
- ii. Helps to manage the data
- iii. Helps to remove the redundancy of data
- iv. Helps scientists to understand the concept of biological phenomena
- v. Provides proper tools to establish new data to existing databases
- vi. Helps the researchers to make new discoveries and achievement in Medicine and Agriculture etc.

3.4 Types of Biological Databases

Based on their contents, biological databases can be roughly divided into two categories:

1. Primary databases

- i. Primary databases are also called as archival database.
- ii. They are populated with experimentally derived data such as nucleotide sequence, protein sequence or macromolecular structure.
- iii. Experimental results are submitted directly into the database by researchers, and the data are essentially archival in nature.
- iv. Once given a database accession number, the data in primary databases are never changed: they form part of the scientific record.

Examples:

- i. ENA, GenBank and DDBJ (nucleotide sequence)
- ii. Array Express Archive and GEO (functional genomics data)
- iii. Protein Data Bank (PDB; coordinates of three-dimensional macromolecular structures).

2. Secondary databases

- i. Secondary databases comprise data derived from the results of analyzing primary data.
- ii. Secondary databases often draw upon information from numerous sources, including other databases (primary and secondary), controlled vocabularies and the scientific literature.
- iii. They are highly curated, often using a complex combination of computational algorithms and manual analysis and interpretation to derive new knowledge from the public record of science.

Examples:

- i. InterPro (protein families, motifs and domains)
- ii. UniProt Knowledgebase (sequence and functional information on proteins)
- iii. Ensembl (variation, function, regulation and more layered onto whole genome sequences).

3. Specialized databases

Specialized databases are databases that are designed to serve a specific research interest. These databases are created with a particular focus on a specific organism or type of data.

Examples of specialized databases include Flybase, the HIV sequence database, and the Ribosomal Database Project.

3.5 Importance of Biological Databases:

Databases act as a store house of information: -

Databases are used to store and organize data in such a way that information can be retrieved easily via a variety of search criteria.

It helps to solve cases where many users want to access the same entries of data.

4. SEQUENCE ALIGNMENT

4.1 INTRODUCTION

One of the most important steps in comparing biological sequences is thought to be sequence alignment. In order to find similarities between two or more nucleotide or amino acid sequences, sequence alignment arranges the sequences. Understanding the functional, structural, and evolutionary links between the sequences is made easier by looking at these areas of commonality. A sequence alignment is a method used in bioinformatics to arrange DNA, RNA, or protein sequences in order to find similar sections between the sequences that could result from functional, structural, or evolutionary ties. Usually, aligned sequences of amino acid or nucleotide residues are shown as rows in a matrix. Gaps are introduced between the residues such that identical or similar characters are aligned in successive columns. Sequence alignments are also utilized for non-biological sequences, including displaying financial data or figuring out how much it costs to move a string in a spoken language.

Sequence comparison is a crucial aspect of bioinformatics analysis that involves comparing newly determined biological sequences with previously known sequences stored in databases.

Sequence alignment is considered the most essential step in comparing biological sequences. Sequence alignment arranges two or more nucleotide or amino acid sequences to identify regions of similarity between the sequences. These regions of similarity are helpful in understanding the functional, structural, and evolutionary relationships between the sequences.

Two commonly used sequence alignment algorithms are global alignment and local alignment.

Frog	G	C	T	T	G	A	C	T	T	C	T	G	A	G	G	T	T
Chicken	G	C	G	T	A	A	C	T	T	C	A	C	A	T	G	A	T
Human	G	C	G	T	C	A	C	T	T	G	A	G	A	C	G	C	T
Rabbit	G	C	G	T	C	A	C	T	T	G	A	G	A	C	G	C	T
Mouse	G	C	G	T	C	A	C	T	T	G	A	C	A	G	G	C	T
Opossum	G	C	G	T	C	A	C	T	T	G	A	G	A	C	G	C	T

Fig: - DNA sequence alignment of different organisms

(Image source: -<http://www.bioss.ac.uk/staff/dirk/talks/lectureDougArm0203.pdf>)

Sequence Alignment then divided into two GLOBAL ALIGNMENTS and LOCAL ALIGNMENT.

Global alignment: Global alignment is a method of comparing two sequences, which aligns the entire length of the sequences by maximizing the overall similarity. This method is used when comparing sequences that are of the same length.

Local alignment: In local alignment, instead of attempting to align the entire length of the sequences, only the regions with the highest density of matches are aligned. This is useful for identifying short conserved regions in protein or nucleotide sequences

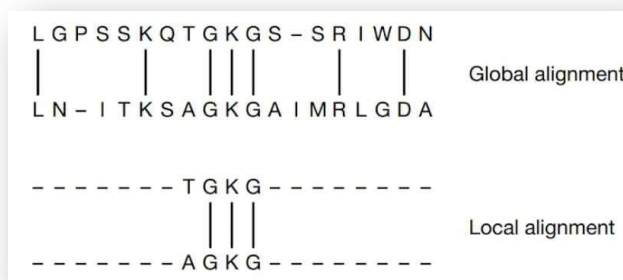


Fig: - GLOBAL & LOCAL SEQUENCE ALIGNMENT

(Image source: <https://microbenotes.com/wp-content/uploads/2023/03/Global-and-Local-Alignment-of-two-sequences.jpg>)

4.2 TYPES OF SEQUENCE ALIGNMENT

1. Pairwise Alignment

- Pairwise sequence alignment is the type of sequence alignment that involves aligning two sequences to identify the optimal pairing of the sequences.
- It is based on a scoring system that assigns positive scores to matching characters and negative scores to mismatching characters or gaps.
- The main objective of pairwise sequence alignment is to obtain the highest possible score, which indicates the degree of similarity between the two sequences.

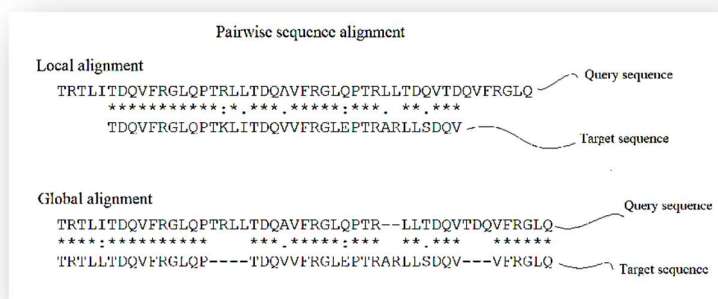


Fig: - PAIRWISE SEQUENCE ALIGNMENT

(Image source: https://bioinformatics.univ-saida.dz/jsbb/March_2023/SAIB_imgs/Tps_Sq_Alns.png)

2. Multiple Sequence Alignment

- Multiple Sequence Alignment involves aligning multiple (three or more) biological sequences to achieve optimal sequence matching.

- Multiple sequence alignments are used to identify conserved sequence regions and to construct phylogenetic trees, which help us understand the functional and evolutionary relationships between different species or groups of organisms.

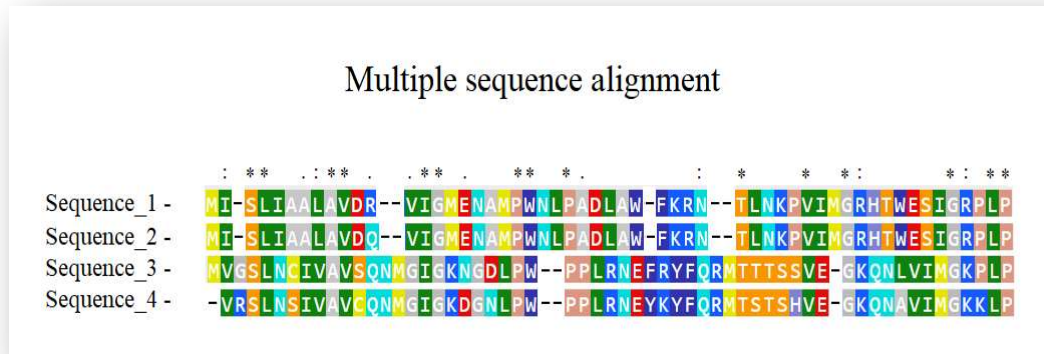


Fig: - MULTIPLE SEQUENCE ALIGNMENT

(Image source: https://bioinformatics.univsaia.dz/jsbb/March_2023/SAIB_imgs/Mlt_Sq_Aln_clr.png)

4.3 TOOLS FOR SEQUENCE ALIGNMENT

I. BLAST

a) INTRODUCTION:

BLAST stands for *Basic Local Alignment Search Tool*. It is a widely used bioinformatics program that was first introduced by Stephen Altschul et al. in 1990 and has since become one of the most popular tools for sequence similarity search. BLAST is a powerful tool for analyzing biological sequence data. Since the initial release of BLAST in 1990, it has undergone continuous updates to improve its speed and accuracy. BLAST is now considered a crucial and widely used tool in the field of bioinformatics. It has played a vital role in numerous research studies and has paved the way for the development of other sequence comparison tools.

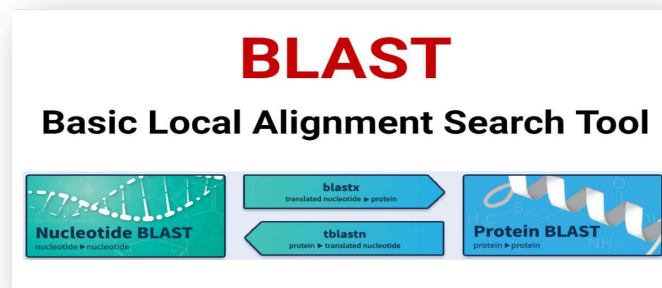


Fig:- Basic Local Alignment Search Tool (BLAST).

(Image source: <https://microbenotes.com/wp-content/uploads/2022/07/Basic-Local-Alignment-Search-Tool-BLAST.jpg>)

II. FASTA

FASTA is one of the first widely-used database similarity search tools. FASTA (or FastA), an abbreviation for 'Fast-All', is a sequence alignment tool that takes nucleotide or protein sequences as

input and compares it with existing databases. It was first developed by David J. Lipman and William R. Pearson in 1985 and has since been refined and adapted for various applications.

The text-based file format for representing nucleotide or protein sequences, which originates from the FASTA program, has now become a standard in bioinformatics. Many other sequence database search tools also use the FASTA file format.

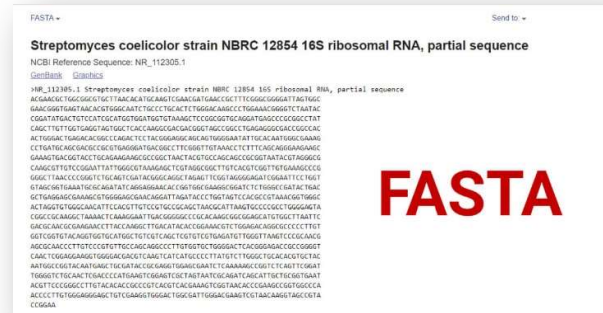


Fig. - FASTA Format.

(Image source: <https://microbenotes.com/wp-content/uploads/2022/07/FASTA-Definition-Programs-Working-Algorithms-Uses.jpg>)

4.4 APPLICATION OF SEQUENCE ALIGNMENT

- Sequence alignment can identify unknown sequences by comparing them with already known sequences in databases.
- Sequence alignment is also used to identify conserved sequence patterns and motifs, which helps to characterize the functions of the sequences.
- Sequence alignment can also produce phylogenetic trees and obtain information about the evolutionary relationship between the sequences aligned.
- Sequence alignment can also predict proteins' secondary and tertiary structures. It can also predict gene locations and new members of gene families.
- Sequence alignment can also be used to develop degenerate PCR primers by analyzing multiple related sequences.

5. PHYLOGENY

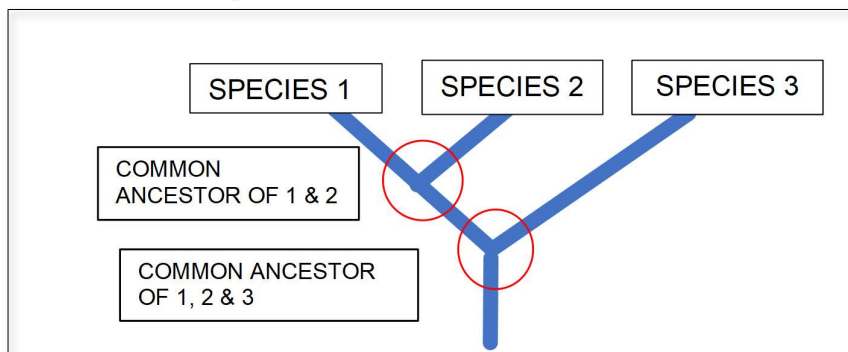
Phylogenetics is the study of the evolutionary history and relationships of organisms, often visualized as phylogenetic trees. These trees represent the branching pattern of descent, showing how different species or groups of organisms are related through shared ancestry. Phylogenetic analysis uses various data, such as genetic sequences or morphological traits, to infer these evolutionary relationships.

4.1 PHYLOGENETIC TREE

A phylogenetic tree (evolutionary tree) is the graphical representation of the evolutionary history of biological sequences and allows us to visualize the evolutionary relationships between them.

Evolution explains the diversity of life on Earth, and understanding evolutionary relationships helps us understand the origin and relationships between different organisms. Molecular phylogenetics is an important aspect of bioinformatics that helps us understand evolutionary history and relationships by using molecular data such as DNA or protein sequences to create phylogenetic trees.

These trees are displayed as two-dimensional diagrams, with branching lines representing the evolutionary history and relatedness of different groups of organisms. The endpoints of the branches represent present-day species or sequences and are referred to as taxa or operational taxonomic units (OTUs). The connecting point where two branches come together is called a node, and it represents an inferred common ancestor of the organisms or sequences connected to the branches. The point where the tree divides into two parts at the bottom is known as the root node, and it represents the common ancestor of all the members of the tree.



5.2 TYPES OF PHYLOGENETIC TREE

- There are several different types of phylogenetic trees. They can be classified as:

On the basis of the presence or absence of a common root:

- Rooted trees** are trees that have a specified root node, which represents the common ancestor of all the organisms in the tree.
- Unrooted trees** do not have a specified root node and show only the branching pattern of the evolutionary relationships among taxa or OTUs, without any information about their common ancestor.

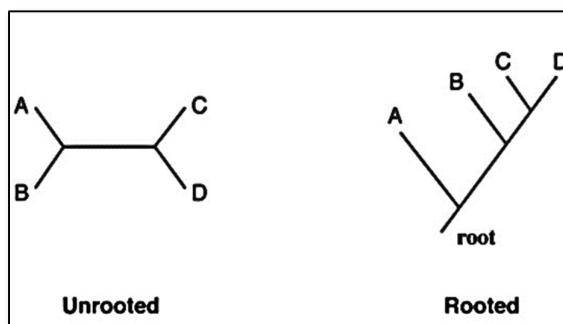


Fig: - TYPES OF PHYLOGENETIC TREE

(Image source: - <https://microbenotes.com/wp-content/uploads/2023/04/Rooted-and-unrooted-Phylogetic-trees.jpg>)

TASK 1

MEDICINAL PLANTS AND HERBS WHICH PLAYS AN IMPORTANT ROLE IN AGING AND OSTEOPOROSIS

ABSTRACT:

Aging happens, and it makes body functions go down slowly, like bone stuff. Osteoporosis is a problem where bones get weak and break more easily, which worries older people a lot, mainly women after menopause. For many years, plants used in medicine helped with bones and aging slowdowns. We look at some important herbs that help keep bones strong and good health overall. The study shows that several medicinal plants and herbs may help prevent osteoporosis and slow my aging, including Epimedium, Moringa officinalis, and Psoralea. To explore the role of medicinal plants and herbs in aging and osteoporosis, a comprehensive review of existing literature and studies is essential. By employing these methodologies, a comprehensive understanding of the role of medicinal plants and herbs in aging and osteoporosis can be achieved, guiding future research and potential therapeutic applications. Studies indicate that plant-derived medicines positively influence bone health by targeting multiple pathways involved in osteoporosis.

KEYWORD:

Anti-aging, Osteoporosis, Bone, Medicinal herbs, Bone Health

INTRODUCTION:

Anti-aging refers to strategies and interventions aimed at slowing, preventing, or reversing the aging process, thereby extending both lifespan and health span—the period of life spent in good health. Aging is a complex, multifaceted process influenced by genetic, environmental, and lifestyle factors. Understanding the underlying mechanisms of aging is crucial for developing effective anti-aging approaches.

The scientific study of aging is known as Gerontology.

Aging is a biological process that results in a progressive and irreversible decline in physical function across all organ systems that is induced by the accumulation of damage in response to a variety of stressors.

In 2013, López-Otín C et al. summarized aging in different organisms (especially mammals) as consisting of nine common features, including genomic instability, telomere attrition, epigenetic alteration, loss of proteostasis, deregulation of nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, and alteration of intercellular communication. [1]

Anti-Aging Strategies: The study shows that there are various measures by which we prevent aging-

- Caloric Restriction and Fasting
- Physical Exercise
- Antioxidant Supplementation
- NAD⁺ Precursors
- Sirtuin Activation
- Skincare Interventions
- Increasing bone density

Lifestyle factors such as a balanced diet, regular exercise, and sun protection remain foundational for promoting healthy aging.

Processes that promote aging and age-related diseases and disabilities which include:

- stress adaptation
- epigenetics
- inflammation
- macromolecular damage
- metabolism
- proteostasis
- stem cells and regeneration.

Aging which raised by following hallmarks:

- their age-related occurrence

- experimentally induced acceleration of aging by enhancing them
- the ability to slow down, halt, or reverse aging by therapeutic modulation on them.

Twelve biomarkers of aging:

- genomic instability
- telomere shortening
- epigenetic changes
- loss of proteostasis
- impaired macro autophagy
- deregulated nutrient sensing
- mitochondrial dysfunction
- cellular senescence
- stem cell depletion
- disturbed intercellular communication
- chronic inflammation dysbiosis.

The World Health Organization (WHO) estimates that by 2050, the world's population of people aged 60 years and older will exceed 2 billion.[2]

These hallmarks are interrelated with one another, as well as with the newly suggested hallmarks of health, consisting of organizational properties of spatial compartmentalization, homeostasis maintenance, and proper responses to stress.

Aging research investigates the declining of organism function in adulthood three conditions that have to hold for every hallmark of aging:

- the time-dependent expression of changes occurring with the aging process
- the potential to accelerate aging by experimentally enhancing the hallmark, and—most convincingly
- the potential to slow down, stop, or reverse aging by therapeutic interventions on the hallmark.[3]

OSTEOPOROSIS:

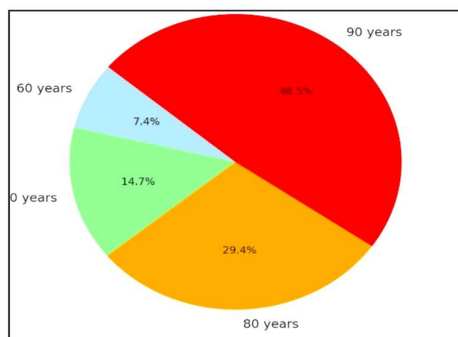
According to WHO report, scientific papers on osteoporosis and aging would probably emphasize the definition of osteoporosis as a disease with low bone mineral density (BMD).

Osteoporosis is a worldwide disease characterized by reduction of bone mass and alteration of bone architecture resulting in increased bone fragility and increased fracture risk. Causes of osteoporosis include increasing age, female sex, postmenopausal status, hypogonadism or premature ovarian failure, low body mass index, ethnic background, rheumatoid arthritis, low bone mineral density (BMD), vitamin D deficiency, low calcium intake, hyper kyphosis, current smoking, alcohol abuse, immobilization, and long-term use of certain medications. The diagnosis of osteoporosis is established by measurement of BMD of the hip and spine using dual energy X-ray absorptiometry.[4]

Osteoporosis is a condition that results in bones that are weak, brittle, and prone to fractures.

According to WHO, Worldwide, osteoporosis is estimated to affect 200 million women - approximately one-tenth of women aged 60, one-fifth of women aged 70, two-fifths of women aged 80 and two-thirds of women aged 90. [5]

OSTEOPOROSIS REPORT AMONG WOMEN BY AGE GROUP



Osteoporosis is a progressive disease that weakens bones, increasing the risk of fractures. It is a chronic and systemic skeletal disease that is defined by low bone mineral density (BMD) along with an increase in bone fragility and susceptibility to fracture. Several medicinal plants have properties that can help slow down aging and support bone health, potentially reducing the risk of osteoporosis.

Osteoporosis is a systemic skeletal disease characterized by a decrease in bone mass and microstructural degradation, which may increase bone fragility and fracture danger and consequently cause serious complications.[6]

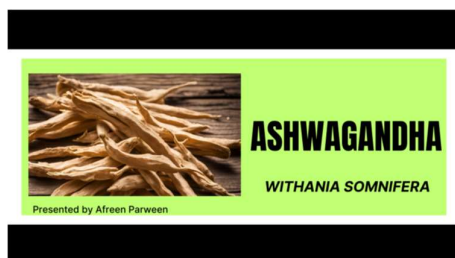
Bone density is a normally distributed variable in the general population, and fracture risk rises continuously with declining BMD. There are two possible approaches to reduce fracture frequency. First, one might attempt to identify and treat a small group of people at the lower extreme of the BMD distribution (a high-risk strategy). Alternatively, one might aim to move the distribution for the entire population (a population-based strategy).[7]

MEDICINAL PLANTS AND HERBS WHICH HAVE THE POTENTIAL TO IMPROVE AGING AND OSTEOPOROSIS:

1. Ashwagandha (*Withania somnifera*):

Rich in antioxidants, helps reduce stress-related aging. Supports bone density and prevents osteoporosis.

- Anti-aging: Rich in antioxidants, helps reduce stress and inflammation, supports collagen production.
- Bone health: Enhances bone mineral density and reduces bone loss.



2. Ginseng (*Panax ginseng*):

Boosts energy, fights aging, and enhances collagen production. Strengthens bones and reduces inflammation.

- Anti-aging: Boosts energy, reduces oxidative stress, and improves cognitive function.
- Bone health: Promotes osteoblast activity, enhancing bone formation. Enhances bone mineral density and reduces bone loss.



GINSENG

PANAX GINSENG

Presented by Afreen Parween

3. Turmeric (*Curcuma longa*):

Turmeric is famous for its bright yellow color and anti-inflammatory properties. The active compound in turmeric, curcumin, has been shown to support bone health by reducing bone loss and inflammation. Adding turmeric to your diet—whether in food, tea, or supplements—can be a great way to keep your bones strong.

- Anti-aging: Curcumin fights oxidative stress, inflammation, and cellular aging.
- Bone health: Helps prevent bone loss and improves bone density.



TURMERIC

CURCUMA LONGA

Presented by Afreen Parween

4. Moringa (*Moringa oleifera*):

Moringa is often called a "superfood" because it's packed with calcium, magnesium, and other essential minerals that strengthen bones. You can add moringa powder to smoothies or take it as a supplement for an easy bone-health boost.

- Anti-aging: Packed with vitamins A, C, and E, which slow aging and improve skin health.
- Bone health: Rich in calcium, magnesium, and phosphorus, essential for strong bones.



MORINGA

MORINGA OLEIFERA

Presented by Afreen Parween

5. Gotu Kola (*Centella asiatica*):

Improves collagen production, reducing wrinkles and sagging skin. Aids in bone repair and strengthens connective tissues.

- Anti-aging: Enhances collagen production, reduces wrinkles, and improves cognitive function.
- Bone health: Supports connective, tissue repair and bone healing.



GOTU KOLA

CENTELLA ASIATICA

Presented by Afreen Parween

6. Shatavari (*Asparagus racemosus*):

Supports hormone balance, reducing age-related bone loss. Strengthens bones and improves calcium retention.

- Anti-aging: Balances hormones and improves vitality.
- Bone health: Supports bone strength, especially in postmenopausal women.



7. Aloe Vera (*Aloe barbadensis miller*):

Aloe vera exhibits potential benefits in addressing signs of aging and supporting bone health, but further research and professional guidance are essential to fully understand and safely harness these effects

- Anti-aging: Hydrates skin, promotes collagen production, and reduces inflammation.
- Bone health: Supports calcium.



In addition, many more plants and herbs which has medicinal properties are mentioned below:

Green Tea (*Camellia sinensis*):

- Anti-aging: Polyphenols like EGCG have strong antioxidant properties.
- Bone health: Helps prevent bone loss and reduce inflammation.



Importance of Natural Product:

Nowadays, the maximum population in developing nations still depends on plant-based medicine that is traditional in nature for their general health care. Herbal medicines are being demanded, and their demand is growing day by day. Herbal plant-based medicines are still popular because they have very low chances of side effects and it is cheap too. It also has a better scope in the international market. The Ayurvedic medicine market is projected to be growing at 20% per annum.

Because of some side effects or ineffectiveness of synthetic drugs, the potential effectiveness of traditional medicines has aroused the interest of scientists and physicians to resort to traditional medicines for the cure of certain chronic and stubborn diseases, such as the cure for osteoporosis. In traditional Chinese medicine (TCM), osteoporosis falls into the category of "rheumatism involving the bone" and "atrophic debility of bones." According to TCM theory of "Kidney dominates bone," numerous medicinal herbs are long used as prescriptions for the treatment of bone metabolic disorders.

The Huang Di Nei Jing (The Yellow Emperor's Classic of internal Medicine) states, "The Meridians move the qi and blood. As a result, tendons and bones get nourished whereas joints get facilitated." Astragalus Bge is among the most widely used herbs for Qi restoration and also can be used for the treatment of osteoporosis. Water extract of A. prevented dexamethasone-induced or ovariectomy-induced osteoporosis in rats by inhibiting osteoclast, reducing bone absorption and enhancing bone formation. In TCM, Curculigo Gaertn., Epimedium Morr., Morinda How., Cistanche salsa G., Eucommia ulmoides Oliv., Psoralea L., Cuscuta Lam., and Dipsacus Mip. have the function of tonifying the kidney, reinforcing Yang, and reinforcing tendons and bones. Conversely, Fructus Ligustri Lucidi (FLL, Chinese name, Nvzhenzi) is one of the herbs of nourishing Yin due to estrogen deficiency. FLL can exert its effect on the pituitary gland and additionally modulate the endocrine function and also have value in prevention of bone marrow loss in patients with cancer on chemotherapy.

The word "Ayurveda" can be described as knowledge (Veda) of the lifespan (ayu). The knowledge is documented in the ancient Indian literature, collectively known as the Veda. Ayurvedic medicine is rooted in breathing exercises, meditation, and yoga. Young bark of **Ficus religiosa** (family-Moraceae), also called Ashwatha or Ashvattha, has been extensively used in Ayurvedic culture for the cure of bone fracture. The stem bark washed with urine of a child (less than 7 years) is used and powdered. Two spoonful of paste were given twice a day for 21 days. Paste is also put on the affected area and bandaged. **Cissus quadrangularis** of family-Vitaceae, which is also called Asthishunkala, is an indigenous Indian medicinal plant. Methanolic extract of Cissus quadrangularis has been found to enhance the healing of experimentally fractured radius-ulna of dogs, as evidenced by radiological and histopathological studies of a young healthy adult, indicating a substantially elevated fracture risk, and how the disease becomes more common with age as a result of the natural decrease in bone mass and remodeling associated with aging. [8]

DISCUSSION:

Aging and osteoporosis are major public health issues worldwide, especially among the elderly, and more so among women. The progressive loss of bone density and overall physiological function highlights the necessity for efficient interventions to control these conditions. Medicinal plants and herbs have been used traditionally as natural remedies to maintain bone health and retard the aging process.

Many medicinal plants, such as Ashwagandha, Ginseng, Turmeric, Moringa, Gotu Kola, Shatavari, and Aloe Vera, have been found to be beneficial in treating both aging and osteoporosis. These plants achieve their therapeutic effects through a variety of biological mechanisms, including lowering oxidative stress, increasing collagen synthesis, activating osteoblasts, and enhancing calcium absorption. Their bioactive molecules provide a safe alternative or complementary method of treatment compared to traditional medical interventions, with fewer adverse effects and greater accessibility, especially in developing countries. The growing worldwide reliance on herbal medicine is the result of its effectiveness, cost, and traditional application in medical systems like Ayurveda and TCM. These systems have long understood the contribution of medicinal plants to longevity and bone strength maintenance. Modern scientific research continues to confirm the efficacy of plant-derived ingredients in improving bone mineral density and minimizing fracture risk. To tap their potential, future studies need to concentrate on strict clinical trials for optimal dosage, action mechanism, and long-term safety of these medicinal plants. Combining herbal medicine with conventional medical practices could promote a more integrated approach to the management of aging and osteoporosis, eventually contributing to better health outcomes for aging populations.

CONCLUSION:

Aging and osteoporosis pose serious health problems, especially among women. Medicinal plants and herbs hold promising natural remedies for preventing bone density loss and promoting healthy aging. Their ongoing study and incorporation into medical practice may improve health results and life quality among aging populations.

TASK 2

METABOLOMICS APPROACHES OF IDENTIFYING POTENTIAL METABOLITES PRESENT IN EFFECTIVE MEDICINAL PLANTS RELATED TO AGING

Afreen Parween

ABSTRACT

The worldwide increase in age-related disorders has accelerated the quest for natural products that can ensure healthy aging. Medicinal plants, which have been used for centuries in traditional medicine, provide a treasure trove of bioactive metabolites with possible anti-aging activities. Metabolomics, a holistic and high-throughput analytical strategy, allows the systematic identification and quantification of these small molecules in complex biological matrices. This method utilizes sophisticated methodologies like mass spectrometry (MS) and nuclear magnetic resonance (NMR) spectroscopy to analyze the metabolite content of medicinal plants. By performing metabolomic studies, scientists are able to identify certain compounds—like polyphenols, flavonoids, alkaloids, and terpenoids—that possess antioxidant, anti-inflammatory, and cell-protective activities, all of which play important roles in preventing age-related cellular injury. Furthermore, metabolomics enables the understanding of metabolic pathways and mechanisms through which these plant-derived metabolites modulate aging processes, such as the regulation of oxidative stress, inflammation, and mitochondrial function. This abstract underscores the importance of metabolomics as a valuable tool in the identification and characterization of potential anti-aging metabolites from medicinal plants towards the discovery of novel therapeutics and nutraceuticals for healthy longevity.

KEYWORD

Aging, metabolomics, metabolites, medicinal plants.

INTRODUCTION

Medicinal plants have long been recognized for their therapeutic properties, including their potential to combat aging and age-related diseases. Recent advancements in metabolomics have allowed for a deeper exploration of these plants, leading to the discovery of specific bioactive compounds that may promote healthy aging. The active constituents of medicinal plants may occur either in roots, leaves, stems, flowers or bark and can be isolated by a suitable extraction process.[1] Metabolomics is a new methodology for analysis of the small-molecule aspect of biological samples and systems. Metabolites form an extremely diverse class of organic small molecules. Metabolomics has already become a valuable “omics” technology that complements genomics, transcriptomics, and proteomics.[2] Metabolomics techniques have significantly improved our understanding of how drug plants can influence aging by identifying bioactive metabolites. Metabolomics is the scientific discipline of chemical reactions in which metabolites, the cell's small-molecule substrates, intermediates, and products, are implicated. Metabolomics is more specifically the "systematic study of the unique chemical fingerprints that specific cellular processes leave behind", the analysis of their small-molecule metabolite profiles.[3]

WHY SCIENTISTS USE METABOLOMICS APPROACHES WHEN STUDYING MEDICINAL PLANTS RELATED TO AGING -

When studying medicinal plants related to aging, scientists use metabolomics approaches to:

- **Identify Active Compounds:** They discover which particular plant metabolites could potentially slow aging or decrease age-related diseases (such as antioxidants or anti-inflammatory substances)
- **Understand How They Work:** Metabolomics assists in uncovering how these plant chemicals interact with human cells and aging pathways (e.g., how they may decrease oxidative stress or enhance cell repair).
- **Compare Different Plants:** By comparing the metabolite profiles of various medicinal plants, researchers can figure out which ones have the strongest anti-aging properties.
- **Quality Control:** It ensures that medicinal plants used in supplements or drugs have the right amounts of beneficial compounds.

ANALYZING INSTRUMENT

Researchers employ equipment such as mass spectrometry (MS) and nuclear magnetic resonance (NMR) to identify and quantify these metabolites.[3] Various combined technologies and methods, including MS-based methods, are used for large-scale investigation of very complex plant extracts. They encompass GC-MS, LC-MS, NMR, MALDI, capillary-based MS, and other MS-based methods.[4] The most commonly used analytical platforms are mass spectrometry, NMR, LC-MS, and GC-MS. These methods allow for extensive data generation and increased chemometric analysis, which yield fundamental information on the metabolites.

IDENTIFYING BENEFICIAL COMPOUNDS

Through metabolomic studies, several compounds in medicinal plants have been linked to anti-aging effects. By contrasting the metabolite profiles of different medicinal plants, scientists can determine which ones possess the most powerful anti-aging effects. These plant compounds are classified into polyphenol (flavonoids, phenolic acids, stilbenes, and lignans), vitamin (vitamins C and A), and terpenoid categories.

- **Polyphenols:** Research shows that plant polyphenols have anti-aging properties. By saying that polyphenols possess 'anti-aging properties,' we imply that these plant-abundant foods contain compounds which can potentially delay or slow down the processes of aging at the cellular and molecular levels. For decreasing aging at cellular level, we consume more polyphenol-dense foods. It also extends your "health span," or the number of healthy years in your life, by reducing blood pressure, cholesterol, and inflammation. Polyphenols have several protective and rejuvenating actions that help in healthier aging and longevity. Polyphenols are fruit and vegetable compounds with antioxidant and anti-inflammatory actions. Certain polyphenols add color and aroma to these foods. There are more than 8,000 varieties of polyphenols, usually flavonoids or non-flavonoids. Polyphenols can be found in many plant foods, like berries, coffee, green tea, and walnuts. Nutrition experts recommend getting polyphenols from plant foods instead of supplements.[5]
- **Vitamin:** Most vitamins cannot be produced in the human body, so they are prescribed through dietary intake.
 1. **Vitamin C:** Vitamin C (L-ascorbic acid or L-ascorbate) is a highly crucial water-soluble antioxidant and arguably the most ubiquitous hydro-soluble vitamin identified to date. The vitamin is suggested for dietary consumption and topical skin treatment because it encourages collagen production in the dermal layer and also plays a part in the resistance to UV-induced injury.
 2. **Vitamin A** can be found in nature in two forms: vitamin A as such, also called retinol, present in a supplementary form in animal feed, and provitamin A, known as carotene, which is found in both animal and plant products.[6]

They then interpret the data to determine which metabolites are present and how they could be of benefit to aging processes. In short, metabolomics helps scientists discover which natural compounds in medicinal plants could promote healthy aging and how they work in the body.

Metabolomics is the technology for the global analysis of metabolites and low-molecular-weight compounds in a biological system, including cells or tissues. Metabolomics facilitates the systematic identification and quantitation of such small molecules engaged in biochemical interactions of organisms and, hence, illuminates complex biological interactions, responses, and functions. The development and engineering of organisms via biotechnology (synthetic biology) demonstrate new biological functions.[7]

Metabolites contribute to biological processes in their role of directly being engaged in the control of physiological processes, like keeping cell homeostasis or transmission of signals through protein–protein interactions. Metabolites are generated and transformed by metabolism of biological system (e.g., cells, tissues, or organisms) including peptides, amino acids, nucleic acids, lipids, carbohydrates, organic acids, vitamins, polyphenols, alkaloids, minerals, and chemical compounds taken in and synthesized by cells or organisms.[8]

Plant metabolites have been employed for thousands of years for medicinal and technological (e.g., preservation, pigments) as well as dietary purposes. With the advancement of technology, it became possible to define the chemical compounds responsible for the bioactivities, leading to the establishment of natural product research, chemo-taxonomy, and numerous other fields. Over the last few years, new chemical analysis technologies became accessible to a broader group of researchers, which led to an unimaginable depth of chemical characterization options previously.[9]

In plants more than 200,000 different metabolites have been reported.

Aging is a natural unavoidable process that is marked by a time-dependent decline of multiple physiological and metabolic functions and is controlled collectively by genetics, proteomics, metabolomics, environment, diet, and lifestyle. The exact mechanism of the aging process is not known, but the metabolomics can contribute substantially by giving a comprehensive metabolite profile and changing metabolomic functions with age.[10]

Aging is characterized by a progressive decline in physiological functions of various organs, followed by dysfunctions, and ultimately death.

According to WHO, In 2020 people aged 60 or above were 1.4 billion. But in 2050 it is estimated to increase the population of people of aged with aged 80 or above will estimated to be triple i.e., approx. 426 million.[11]

Aging and age-related diseases, such as cardiovascular disease, cancer, diabetes, and neurodegenerative disorders, which have grown exponentially, are inherently associated with redox imbalance and oxidative stress. There are hundreds of biologically active constituents in each whole plant matrix, which offer promissory bioactive effects to human beings. In fact, the global population has given more attention and preference for the utilization of medicinal plants for healthy aging and longevity promotion.[12]

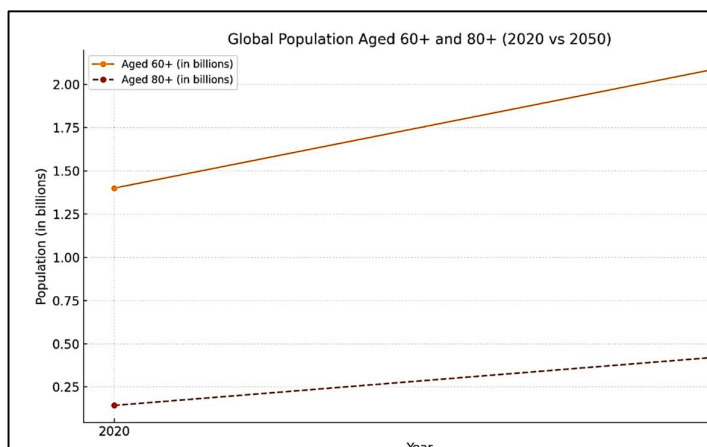


Fig.1 Here's the graph illustrating the projected global population growth of people aged 60+ and 80+ between 2020 and 2050.

The effective medicinal plants which are beneficial for aging:

There is a variety of herbs that have been traditionally employed by people for aging. The herbs commonly identified as medicinal plants are widely known for their healing properties and are used traditionally to cure various diseases, including aging. It includes herbs like **Ashwagandha, Turmeric, Moringa oleifera, Amla, Shatavari, Aloe vera.**

1.ASHWAGANDHA:

Ashwagandha or Indian ginseng is also known as Indian winter cherry, Ashwagandha, or the herb *Vitania sluggard* (*Withania somnifera*). The raw material used in medicine is the root, and the name "Ashwagandha" is derived from the word "ashwa", meaning horse. It is believed that after consuming the root, one gains powers similar to that of a horse. The second part of the name "gandha," means fragrance and refers to the characteristic smell of the fresh root of the plant. Since ancient times, it has been traditionally used in Ayurvedic medicine as a substance that strengthens the nervous system. This is evidenced by its adaptogenic effects and medicinal use of the so-called "rasayana". Ashwagandha is botanically called *Withania somnifera* Dunal and is a member of the Solanaceae family commonly called Indian Ginseng or Winter Cherry. The name "Ashwagandha" literally means "smell of horse" since the herb's fresh roots smell like horse and the fact that one who takes the herb extracts may develop the vitality and strength of a horse. It has multifarious rejuvenative effects on the human body and thus.[13,14]

Metabolites present in ashwagandha are:

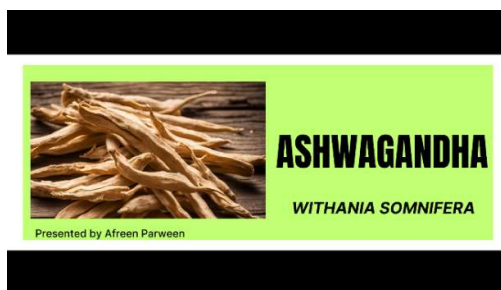
It possesses a diverse array of bioactive metabolites responsible for its therapeutic activity. The major categories of these metabolites are withanolides, alkaloids, and saponins.

Withanolides: These are steroidal lactones which are specific to *W. somnifera*. Some withanolides like withaferin A have shown an array of biological activities, including anticancer and anti-inflammatory activities. More than 40 withanolides have been found in ashwagandha by studies.

Alkaloids: Ashwagandha is composed of several alkaloids, such as anahygrine, berberine, and Cusco hygrine. These alkaloids occur predominantly in the roots of the plant and are responsible for its medicinal actions.

Saponins: These, along with withanolides and alkaloids, are some of the active ashwagandha constituents.

The various metabolites found in ashwagandha are accountable for its broad spectrum of medicinal uses, ranging from stress reduction, immune function, and anti-inflammatory activity.



2. TURMERIC: As known as “Golden spice”. Its scientific name is “*Curcumina longa*”. Turmeric has been in use since almost 4000 years ago during the Indian Vedic civilization, when it was employed as a spice in cooking and possessed religious meaning as well. It likely was introduced in China by 700 ADS, in East Africa by 800 ADS, in West Africa by 1200 AD, and in Jamaica in the eighteenth century.

Turmeric is a herb whose medicinal history spans a very long time, stretching almost 4000 years into the past. Turmeric, in Southeast Asia, serves not just as a major spice but also as a ritualistic component. Turmeric is sometimes referred to as "Indian saffron" due to its brilliant yellow hue.

Different names in different cultures

Turmeric is today cultivated all over the tropics and known by various names in various countries and cultures. It is well known as "haldi" in North India, a term that is traced back to the Sanskrit word *haridra*, and is referred to as "manjal" in the south, a term that was often used in ancient Tamil poetry. The origin of the name turmeric is from the Latin *terra merita* (meritorious earth), which relates to the mineral pigment-like color of ground turmeric. Turmeric is known as *terre merite* in French and as simply "yellow root" in a number of different languages. Its name in a number of different cultures is based on the Latin word *curcuma*. In Sanskrit, turmeric is named at least 53 different ways, among them *anestha* (not provided for homa or sacrifice), *Bhadra* (fortunate or auspicious), *bahula* (abundance), *dhirgharaja* (long in shape), *gandhaplashika* (one which yields pleasant smell), *Gauri* (makes fair), *gharshani* (rubbing), *haldi* (that invites attention with its color), *haridra* (beloved of Hari, Lord Krishna), *harita* (greenish one), *hemaragi* (demonstrates golden color), *hemaragini* (provides golden color), *hridayavilasini* (delights the heart, charming), *Jayanti* (conqueror of diseases), *jawarantika* (fever destroyer), *Kanchani* (demonstrates golden color), *Kaveri* (harlot), *krimighni* or *kashpa* (destroyer of worms), *kshamata* (ability), *Laxmi* (good fortune), *mangalprada* (one who provides auspiciousness), *mangalya* (auspicious), *Mehagni* (fat destroyer), *Nisha* (night), *nishakhya* (is known as night), *nishawa* (clears darkness and provides color), *patwaluka* (perfumed powder), *Pavitra* (holy), *Pinga* (reddish-brown one), *pinja* (yellow-red powder), *pita* (yellow one), *pitika* (which provides yellow color), *rabhangavasa* (which destroys fat), *ranjani* (which provides color), *ratrimanika* (as lovely as moonlight), *shifa* (fibrous root), *shobhna* (dazzling color), *shiva* (noble), *shyama* (dark), *soubhagaya* (fortunate), *survana* (golden color), *survanavara* (which displays golden color), *tamasini* (as lovely as night), *umavara* (Parvati, Lord Shiva's wife), *vairagi* (who remains unattached to desires), *varavarnini* (which gives fair complexion), *varna datri* (one who improves body complexion), *Yarnini* (which provides color), *vishagni* (destroyer of poison), *Yamini* (night), *yoshitapriya* (wife's lover), and *yuvati* (young girl).

In Ayurvedic medicine:

Turmeric is a well-documented treatment for various respiratory conditions (e.g., asthma, bronchial hyperreactivity, and allergy), as well as for liver disorders, anorexia, rheumatism, diabetic wounds, runny nose, cough, and sinusitis. Turmeric (*Curcuma longa*) contains a diverse array of bioactive metabolites, each contributing to its characteristic properties and potential health benefits.^[15]

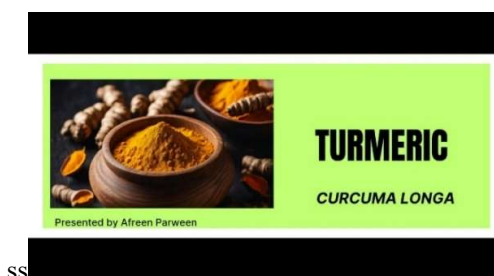
The primary metabolites found in turmeric:

Curcuminoid: Curcuminoids are polyphenolic chemicals that give turmeric its yellow color and are mostly known for their antioxidant and anti-inflammatory activity. The key curcuminoids are:

- **Curcumin:** The dominant curcuminoid, which makes up the majority of the curcuminoids in turmeric.
- **Desmethoxycurcumin:** A curcumin derivative lacking one methoxy group, contributing to the bioactivity of turmeric.
- **Bisdemethoxycurcumin:** A second curcumin derivative missing two methoxy groups, which also contributes to the medicinal benefits of the spice.

Turmeric's essential oils include turmerone, germacrone, atlantone, and zingiberene, among others.

These compounds collectively contribute to turmeric's characteristic properties and potential health effects.



1. **MORINGA:** *M. oleifera* known as “miracle tree” is increasingly used in nutraceuticals for the reported health effects and nutritional value of its leaves.[16] *Moringa oleifera* (*M. oleifera*), commonly employed in tropical and subtropical areas, has been said to exhibit excellent anti-aging effects. *Moringa oleifera* (*M. oleifera*) is native to India, and it is widely planted and used in tropical and subtropical areas in recent decades, due to its high nutritive values in proteins, amino acids, fats, minerals, and vitamins. On the other hand, the complex and varied chemical constituents such as flavonoids, phenolic acids, glucosinolates, and nitrile glycosides in *M. oleifera* result in its many pharmacological activities, such as antioxidant, anti-inflammatory, antibacterial, anticancer, hypoglycemic and hypolipidemic activities.[17]

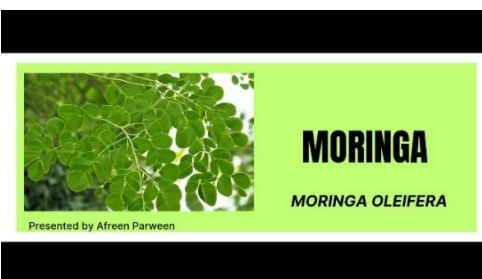
Metabolites present in *Moringa oleifera*:

Moringa oleifera, commonly known as the drumstick tree, contains a diverse array of bioactive metabolites contributing to its nutritional and therapeutic properties. *Moringa oleifera* improves immunity and prevents disease. A plant rich in antioxidants and anti-inflammatory compounds, moringa exhibits potential anti-cancer capabilities in addition to its nutritional value. The antidiabetic, anticancer, lung-protective, and hepatoprotective activities (e.g., antioxidant activities) of extracts of various parts of *M. oleifera* can be attributed to reference to the range of phytochemicals present in the plant, including flavonoids, glucosinolates, sterols, terpenoids, thiocarbamates, and phenolics.[18]

Flavonoids and Phenolic Acids:

- **Flavonoids:** These are strong antioxidants found in *Moringa oleifera* leaves.
- **Phenolic Acids:** These acids are responsible for the antioxidant activity of *Moringa oleifera*.
- **Carotene:** It is a type of pigment that is converted into vitamin A in the body. It is known for its antioxidant properties and various health benefits. The leaves of the moringa plant are rich in minerals like magnesium, calcium, potassium, phosphorus, Sulphur, iron, and copper as well as fiber, fat, and protein.

These metabolites contribute in aggregate to the nutritional quality and possible health effects of *Moringa oleifera*.



4.AMLA: *Phyllanthus emblica* (syn. *Emblica officinalis*), commonly known as the Indian gooseberry, is an important medicinal plant in the traditional Indian system of Ayurvedic medicine. The fruits, which are of both dietary and medicinal use, are shown to possess myriad medicinal benefits and to possess anti-aging effects. Amla has properties that can fight age-related diseases like cancer, diabetes, cardiovascular disease, renal failure, immunosuppression, arthritis, cataract, and skin wrinkling. Amla branches were standardized and examined the anti-skin aging activities.

- Sinapic and ferulic acids were dominant phenolics in extract.
- Cellular and in vitro assays showed the matrix metalloproteinase-2 inhibition and melanogenesis as well as antioxidants.
- Clinical research confirmed the lightening of skin color, the improved skin elasticity and hydration, and reduction in skin wrinkles.
- Research showed that amla branch is a source of bioactive compounds for use as an anti-skin aging ingredient.[\[19\]](#)

Amla is a rich source of many bioactive metabolites that impart its nutritional and therapeutic potential.

Polyphenols:

- **Ellagitannins:** Amla is found to contain elevated concentrations of ellagitannins, such as emblicanin A (37%), emblicanin B (33%), punigluconin (12%), and pedunculagin (14%), that impart antioxidant activities.
- **Flavonoids:** Flavonoid quercetin and kaempferol, characterized by their antioxidant activity, are significant flavonoids found in amla.

Organic Acids:

- **Ascorbic Acid (Vitamin C):** Amla is also known for its high concentration of vitamin C, which adds to its antioxidant activity.
- **Gallic and Ellagic Acids:** These phenolic acids are responsible for the antioxidant and anti-inflammatory activity of amla.

Fatty Acids:

- **Linolenic, Linoleic, Oleic, Stearic, Palmitic, and Myristic Acids:** These fatty acids are found in amla, adding to its nutritional value.

Other Phytochemicals:

- **Tannins, Saponins, and Glycosides:** These secondary metabolites occur in amla and are responsible for its medicinal value.

These metabolites all contribute to the health benefits of amla, such as antioxidant, anti-inflammatory, and lipid-lowering activity.



5. SHATAVARI: Shatavari (*Asparagus racemosus* Willd) has been employed as an Ayurvedic women's health herb for centuries, but there is limited empirical support for its efficacy. Steroidal saponins are considered the major bioactive components of the root of shatavari. The saponins of shatavari are called shatavarins I–IV, which are sarsasapogenin glycosides. The root of shatavari also includes other chemical constituents such as racemosides, racemose, racemofuran and asparagamine A, which exhibit antioxidant activity.

Shatavari administration in rats induces pro-estrogenic changes in mammary glands and genital organs.

The estrogen-like potential of shatavari supplementation has relevance to both postmenopausal muscle and bone well-being. Observational evidence has long indicated that in women, post-menopausal estrogen deficiencies may favor the age-related reduction in muscle force and function (sarcopenia); men show a gradual decline in handgrip strength (HGS) over the lifespan, while women show a precipitous decline in such strength around mid-sixth decade

In Ayurveda, Shatavari is used to balance Pitta and Vata dosha, though it may increase Kapha if taken in excess. Shatavari, the miracle herb with a hundred roots has been traditionally associated with boosting female health and balance during all stages of life.

Benefits of Shatavari:

- Anti-inflammatory benefits from the presence of racemofuran.
- Adaptogenic properties help to withstand stress and improve the mood.
- Antioxidant properties to prevent free radical damage.
- Promoting stronger immunity by building and replenishing tissues.[20,21]

Metabolites present in Shatavari:

Shatavari (*Asparagus racemosus*) is a medicinal plant widely used in Ayurveda for its adaptogenic, immunomodulatory, and reproductive health benefits. Its bioactive metabolites include steroidal saponins, flavonoids, alkaloids, and polysaccharides, which contribute to its therapeutic properties.

Metabolites of Shatavari

Steroidal Saponins: Exhibit adaptogenic, anti-inflammatory, and reproductive health benefits.

- Shatavari (Shatavarin I–IV)
- Asparagoides
- Racemosides
- Sarsasapogenin

Flavonoids: Act as antioxidants and contribute to anti-inflammatory properties.

- Quercetin
- Rutin
- Kaempferol
- Isoflavones

Alkaloids: Possess neuroprotective and antimicrobial effects.

- Asparagamine A

Polysaccharides & Oligosaccharides: Enhance immune function and act as prebiotics.

- Immunomodulatory polysaccharides
- Fructooligosaccharides

Phenolic Compounds: Provide antioxidant activity and contribute to anti-aging effects.

- Ferulic acid
- Caffeic acid



6.ALOE VERA (*Aloe barbadensis* Miller): Aloe vera is a tropical cactus that's been used for centuries to heal the skin from a number of different ailments. Nowadays, because of its healing and moisturizing properties, aloe vera is commonly used as an ingredient in all sorts of skin care products. Aloe vera is also used by many to minimize the presence of wrinkles and fine lines.

Aloe Vera and it is a material on which there have been allegations made regarding its anti-inflammatory, healing, moisturizing, antibacterial, antifungal and antiviral attributes. The gel is extracted from the pulp of aloe vera, a tropical cactus plant of the lily family, and aloe was used as folk medicine since Roman times and it is now a well-known ingredient in a broad variety of healthcare and cosmetic products. The gel consists of the inner, transparent portion of the aloe vera leaf, and the exudate from the outer tissues is also employed therapeutically.[22,23]

Metabolites present in Aloe Vera are:

It contains several bioactive metabolites that contribute to its anti-aging properties by promoting skin hydration, collagen synthesis, and antioxidant defense.

- **Polysaccharides** (Acemannan):
Stimulates collagen synthesis, increases skin hydration, and promotes wound healing.
Anti-Aging Benefit: Enhances skin elasticity and decreases wrinkles.
- **Anthraquinones** (Aloe-emodin, Aloin):
Have antioxidant and anti-inflammatory activity.
Anti-Aging Benefit: Shields skin from oxidative stress and retards cellular aging.
- **Vitamins (A, C, E)**
Have strong antioxidant activity.
Anti-Aging Benefit: Scavenges free radicals, inhibiting oxidative damage to skin cells.
- **Phenolic Compounds**
Have strong antioxidant activity.
Anti-Aging Benefit: Shields skin from photoaging due to UV exposure.
- **Sterols (Lupsterol, Campesterol, β -sitosterol):** Anti-inflammatory and skin barrier-enhancing activity.
Anti-Aging Benefit: Enhances skin hydration and decreases skin irritation.
- **Amino Acids:** Provide necessary building blocks for protein construction.
Anti-Aging Benefit: Facilitate skin repair and enhance firmness.
- **Enzymes** (Brady kinase, Catalase)
Inhibit inflammation and scavenge free radicals.
Anti-Aging Benefit: Decrease skin irritation and counteract oxidative stress.



DISCUSSION

The increasing global health burden of age-related diseases has spurred the quest for natural products promoting healthy aging. Medicinal plants, highly valued in traditional medicine for centuries, present a vast source of bioactive metabolites with potential anti-aging activities. Metabolomics, a holistic and high-throughput analytical approach, has transformed our capability to systematically screen and quantify these metabolites, unveiling a deeper understanding of how medicinal plants can modulate aging processes. Metabolomics utilizes cutting-edge technologies such as mass spectrometry (MS) and nuclear magnetic resonance (NMR) spectroscopy to screen complex biological matrices. Such technologies allow the identification of bioactive small molecules—flavonoids, polyphenols, alkaloids, and terpenoids—which possess antioxidant, anti-inflammatory and cytoprotective activities critical in reversing age-related cellular injury. Profiling these bioactive compounds by scientists can decipher the intricate metabolic pathways that lead to aging and age-related diseases.

The use of metabolomics in aging studies has various benefits. To begin with, it facilitates the identification of particular metabolites that are capable of slowing down the aging process or reversing age-related ailments. Polyphenols, for example, are well known to possess antioxidant activity, which enables them to neutralize ROS and alleviate oxidative stress—another key factor in aging. Quercetin and epigallocatechin gallate, for example, have exhibited considerable protective activities against cellular aging.

In addition, metabolomics enables the comprehension of how these bioactive molecules interact with human cells and affect key aging pathways. For instance, some plant metabolites regulate mitochondrial function, inflammation, and DNA repair mechanisms. This integrated understanding is crucial for the design of targeted therapeutics and nutraceuticals for healthy longevity.

Another important advantage of metabolomics is its application in quality control. Through detailed metabolite profiling, scientists are able to confirm the consistency and effectiveness of medicinal plant-derived supplements and medicines. This is of particular importance for the translation of traditional medicinal wisdom into scientifically tested and standardized drugs.

Research on individual medicinal plants has identified the rich variety of bioactive metabolites with anti-aging activities:

Aloe vera: Rich in polysaccharides and phenolic compounds, Aloe vera is characterized by antioxidant and anti-inflammatory properties, thus showing promise in improving the health of skin and postponing aging.

Ashwagandha (Withania somnifera): This is an herb with withanolides and alkaloids. Ashwagandha possesses adaptogenic activity, fighting against stress, which causes oxidative damage.

Turmeric (Curcuma longa): The most significant curcuminoid of turmeric, curcumin, shows intense antioxidant and anti-inflammatory effects and, thus, acts as a potent anti-aging agent.

Moringa (*Moringa oleifera*): Rich in flavonoids, glucosinolates, and isothiocyanates, which are responsible for its antioxidant and anti-inflammatory activities.

Amla (*Phyllanthus emblica*): Rich in vitamin C and polyphenols, amla aids collagen formation and oxidative stress protection.

Shatavari (*Asparagus racemosus*): Steroidal saponins present in shatavari have antioxidant activity and maintain hormonal balance, making it useful for anti-aging.

Knowledge of the metabolic pathways affected by these plant-derived metabolites is informative about their mechanisms of action. For example, polyphenols are able to activate cellular antioxidant defenses via the Nrf2 pathway, whereas flavonoids can suppress pro-inflammatory signaling via the NF- κ B pathway. These interactions highlight the intricate network of biochemical processes that can be revealed by metabolomics.

CONCLUSION

Metabolomics is a strong tool in the discovery of medicinal plants for anti-aging uses. Through facilitating the holistic examination of bioactive metabolites and their interactions with biological systems, metabolomics opens the door to the creation of new therapeutics and nutraceuticals to support healthy aging and longevity. This holistic strategy not only integrates traditional knowledge and contemporary science but also has the potential to combat the worldwide issue of age-related diseases.

SECTION B
RESEARCH PROJECT

INTEGRATIVE MOLECULAR DOCKING APPROACHES IN TARGETING GENOMIC ALTERATIONS FROM WHOLE GENOME & EXOME SEQUENCING IN PANCREATIC CANCER

ABSTRACT: Pancreatic cancer is the 12th most common cancer worldwide, with approximately 510,992 new cases reported in 2022. The overall prognosis for this affliction is further impaired by the stark reality that merely 6% of patients endure a long-term survival. Surgical excision of the offending tumor is the singular curative intervention available, yet unfortunately, only approximately 20% of cases are diagnosed at a stage sufficiently early to facilitate operative intervention, underscoring the imperative necessitating the development of enhanced detection and therapeutic modalities. Pancreatic cancer originates in the tissues of the pancreas, an organ that plays a crucial role in digestion and blood sugar regulation. Genomic alterations play a crucial role in the development and progression of pancreatic cancer, particularly in understanding its underlying mechanisms and potential treatment strategies. This review seeks to assess integrative molecular docking methods aimed at genomic changes detected by whole genome and exome sequencing in pancreatic cancer, their potential for improving personalized treatment options, and patient outcomes in this deadly disease.

KEYWORD: Genome alterations, Pancreatic cancer, Whole Genome Sequencing, Whole exome sequencing, Molecular Docking, Fisetin, Luteolin.

1. INTRODUCTION: Pancreatic cancer is a type of cancer that begins as a growth of cells in the enzymes that help manage blood sugar. It represents a significant challenge in oncology due to its aggressive nature and poor prognosis. Approximately 90% of cases are Pancreatic Ductal Adenocarcinoma (PDAC). In early stages it's asymptomatic. Due to its late diagnosis, it becomes the deadly disease. It is the type of cancer that begins as a growth of cells in the enzymes that help manage blood sugar. [1,2] According to cancer statistics 2024, estimated new cases of Pancreatic cancer according to gender in U.S. According to these new cases in both gender is 66,440, in male 34,530, female 31,910. Estimated deaths of both gender is 51,750, Male 27,270 and Female 24,480. Based on updated estimates from International Agency for Research on Cancer (IARC), the global pancreatic cancer statistics by world region in year 2022, new cases is 510,5666. Death cases- 467,005. [3,4] Treatment for PDA which is a most common type of PC, becomes more complex due to resistant against targeted therapies, immunotherapies, chemotherapy & radiation. [5] The major modifiable risk factors are tobacco smoking, obesity, diabetes, and chronic pancreatitis, whereas the non-modifiable risk factors are age (predominantly >65 years), male sex, race (increased in African Americans), family history, and inherited syndromes [6] In this review, we discuss about the integration of genomic alterations by analysis of WGS & WES in Pancreatic cancer using Molecular docking through various types of bioinformatics tools. Genomic Alteration in Pancreatic Cancer Whole genome analysis, both coding, non-coding, and mitochondrial DNA, Identification of novel genomic variants (structural, single nucleotide, insertion-deletion, copy number), Detection of previously unidentified variants for future research targeting. [7] Whole genome sequencing in Pancreatic cancer redefines the mutational landscape. [8] It identifies rare genetic variants in familial pancreatic cancer patients. [9] Whole exome sequencing, also known as whole exome sequencing (WES), is a genomic technique for sequencing all of the protein-coding regions of genes in a genome (known as the exome). [10] Whole-exome sequencing of pancreatic cancer defines genetic diversity and therapeutic targets. [11] This review seeks to discuss integrative use of molecular docking methods with genomic findings of WGS and WES data in pancreatic cancer. It emphasizes the use of these integrated methodologies to recognize druggable targets, design structure-based drugs, and bridge existing gaps in therapeutic effectiveness. The review critically assesses existing achievements, case studies, and translational issues in the process of advancing docking-derived therapies from the bench to the bedside in the treatment of genetically defined subtypes of PDAC.

2. MATERIALS AND METHOD

2.1 DATA COLLECTION AND META ANALYSIS

The data for this study was collected from publicly available NCBI- Gene Expression Omnibus (GEO) dataset. The target populations of study were the patients who had Pancreatic cancer. In this study seven cases were taken. The selected datasets were categorized on the basis of test and control are GSE180971, GSE288408, GSE263236, GSE223407, GSE183795, GSE140624, GSE172168. Re-analysis of the collected data was done and the result was further sorted on the basis of p.value <0.05. Further, this data was analyzed by using Venn Master software to find out the commonly expressed genes among all these seven cases. The highly conserved genes among all the targeted disease were analyzed by Cytoscape for the identification of hub genes.

2.2 HOMOLOGY MOLDELLING AND PHYSIOLOGICAL CHARACTERISTICS

PPP2CB was taken as taken as receptor for homology modelling. UNIPROT was used to impact the sequences for both the genes. In continuation to this for evaluating physiochemical properties ProtParam tool was used and secondary structure prediction by SOPMA. And also, transmembrane properties were noted down through SOSUI server and Ramachandran plot was downloaded through Ubuntu (an operating system). Then the PPP2CB genes was prepared as receptor by removing the existing ions and final receptor hydrogen and charge were added and saved as PDB format for further analysis.

2.3 RECEPTOR PREPARATION

The protein files were extracted from Swiss model were forwarded for receptor preparation using UCSF Chimera 1161 Receptor preparation with Chimera was preceded further with the removal of residues such as ions, ligands and solvents from protein. Finally, in receptor preparation pipeline chimera is again used to carry the energy minimization of the protein then surface binding analysis is done and save the file as final receptor in default .pdb format.

2.5 LIGAND PREPARATION

Extensive research literature study suggested us to explore various phytochemical compounds useful in treatment of this disease. We screened and downloaded various compounds from PUBCHEM present then library preparation. With the help of properties of bio compounds, OSIRIS property explores helps to design structures and calculates numerous drug-relevant characteristics to be reasonable for candidate drug. Prediction results are valued and colour coded. Properties with high prospects of unwanted results like mutagenicity or green colour indicates drug behavior. Drulito (Drug-likeness tools) rules are set of guidelines for the structural properties of compounds, used for fast calculation of drug-like properties of a molecule. These guidelines are not absolute, nor are they intended to form strict cutoff values for which property values are drug-like and which are tint drug-like. Nevertheless, they can be quite effective and efficient design the chemical shown in red whereas.

2.5 MOLECULAR DOCKING

Molecular docking is a key tool in structural molecular biology and computer-assisted drug can be used to perform design. The goal of ligand-protein docking is to forecast the foremost binding models) of ligand with a protein of known three- dimensional structure. Docking studies were carried out using CHIMERA-Auto dock vina. Docking score with high negative value represent high binding energy score AG that indicates the principle of minimum energy and high stability leading to best interaction between proteins and the ligands. The ligand-receptor complex that we docked were further visualized in Biovia Discovery studio so that we can interpret the docking analysis between the bonds of ligands and receptors.

Further downstream analysis were carried out to check the metabolic pathways for both common & unique genes & based on literature most relevant genes were selected for target prediction. However in certain cases top 20 hub genes metabolic pathways ranking was little less than the selected genes which were used for target prediction.

The potential top 20 hub genes can be also cross validated for target prediction in coming duration.

3. RESULTS

3.1 GENOMIC AND META ANALYSIS OF SAMPLES

After going through various literatures, research articles regarding Pancreatic cancer, total 7 cases were selected for the further research work (meta- analysis). All the raw data of selected cases were downloaded after the recanalization with GEO2R. All the samples were subjected to read normalization prior to genomic analysis having (p-value 0.05) for the test and controlled and also the condition cases samples of Pancreatic Cancer cases resulted in the identification of total -represented as Box plot (Figure 3.1), Mean difference plot (Figure 3.2) and Volcano plot (figure 3.3). The qualified set of significance genes across the seven cases were imported for gene conservancy analysis in order to highlight the most commonly expressed set of genes present in seven diversified cases of Pancreatic cancer. Further we identified top 10 hub genes from which PPP2CB was taken for further research.

3.2 TARGET RECEPTOR PREPARATION

After the selection of most commonly expressed genes i.e., PPP2CB due to their close relatedness with Pancreatic cancer, homology modelling of the gene was carried out which start from the ProtParam in which Physiochemical properties of the PPP2CB was studied, shown in (Figure 3.5) and showed instability index of 32.92, hydrophobicity (GRAVY) of -0.446. The genes were further analyzed to check secondary structures analysis of a protein and detection of helix which resulted in identification of length of PPP2CB as 309 and similarity threshold value as 8 as shown in (Figure 3.6 & 3.7) using SOPMA & SOSUI which are an open online platform. Ramachandran Plot of PPP2CB was plotted showing favoured regions 83.06 and additional allowed region 16.94 as shown in (Figure 3.8). The PPP2CB potential gene was then imported for receptor preparation to UCSF Chimera in which additional chains, ions, solvents and ligands were removed. Then by using the SPDV, the receptor was carried out for the energy minimization through GAFF force field for 5000 steps. Finally for the hydrogen and charge were added, leads to the final preparation of receptor.

3.3 LIBRARY PREPARATION OF BIO-ACTIVE COMPOUNDS

A total of 2 metabolites were selected from herbal plants to study their medicinal properties to cure Pancreatic cancer through several research and review papers further the metabolites were passed through several research and review papers further the metabolites were passed through Drulito and Osiris to check their ADMET properties wherein metabolites were filtered on parameters like cLogP, solubility, molecular weight, TPSA, drug likeliness and drug score. (Table 1 & 2) both ligands passed all screening process showing no toxicity such as irritant, mutagenic, tumorigenic and reproductive effective. Both these ligands named Fisetin and Luteolin were selected for final ligand preparation.

3.4 LIGAND SCREENING

Both Fisetin and Luteolin cleared toxicity filters and exhibited perfect drug-likeness scores. Their physicochemical properties (e.g., molecular weight <500 Da, cLogP within range) and structural profiles (e.g., correct number of bonds, angles, and torsions) further supported their viability for molecular docking.

3.5 DOCKING OUTCOMES

Total 2 ligands with 1 potential target receptors were selected for molecular docking studies. On the basis of values for pocket ID for PPP2CB as X= -4.985, Y= -3.714, Z= 6.262 with size 20 in center. The Binding energy (AG) of molecular docking of PPP2CB with Fisetin is -5.630 and of PPP2CB with Luteolin is -5.145. These negative values of binding energy refer to thermodynamically stable and favorable interactions that indicate strong ligand–receptor affinity under present conditions of docking. The ligands established important hydrogen bonds and hydrophobic interactions in the PPP2CB binding site that suggest potential biological activity and stability of interactions.

4. DISCUSSION:

In this study, we have shown an integrative analysis of whole genome and exome sequencing data in pancreatic cancer with known mutations using molecular docking. By implicating the genomic findings, we found several extremely conserved genes and among them PPP2CB was a central hub gene that plays a critical role in pancreatic carcinogenesis. Homology modeling and receptor preparation were carried out using sophisticated bioinformatics tools to create a stable structural platform for conducting docking simulations.

Fisetin and Luteolin were chosen as phytochemicals based on their observed bioactivity, local availability in Jharkhand's medicinal plants, and modeled toxicity profiles. Toxicity screening with the Osiris Property Explorer, however, indicated that Fisetin has a mutagenic alert (red flag) where there is a perceived risk of genetic toxicity in biological systems. Nevertheless, docking studies indicated that both compounds had good binding affinities with PPP2CB with high negative binding energies predicting thermodynamically stable and effective interactions.

These findings validate the potential biological activity of both ligands, though Fisetin's mutagenic issue should be thoroughly assessed by way of additional in vitro and in vivo toxicity studies prior to moving forward as a therapeutic candidate. Luteolin, with its less troublesome toxicity profile, is promising for potential future therapeutic use.

This integrative strategy fills the gap between genomic studies and clinical practice. WGS/WES-derived data-guided docking provides a precision oncology platform that can facilitate drug discovery and personalization of therapy for pancreatic cancer.

5. CONCLUSION:

This research illustrates the power of combining whole genome and exome sequencing information with molecular docking in order to discover and validate therapeutic targets for pancreatic cancer. The discovery of PPP2CB as a central hub gene and its consistent interactions with phytochemicals such as Luteolin and Fisetin illustrates the potential of natural compounds for targeted therapy. Although both compounds had high binding affinities, Fisetin's Osiris mutagenic alert for the toxicity test is concerning and requires additional toxicological confirmation. Luteolin's safer profile makes it a better choice. This strategy promotes precision oncology by allowing targeted, genome-guided drug discovery for the treatment of pancreatic cancer.

6. ACKNOWLEDGMENT:

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3.8 CONFLICT OF INTEREST: There is no conflict of interest in the presentation of paper.

3.9 FUNDING SOURCE: There was no funding source.

RESULTS

GRAPHS OBTAINED FROM META – ANALYSIS OF THE GEO DATASET GSE288408

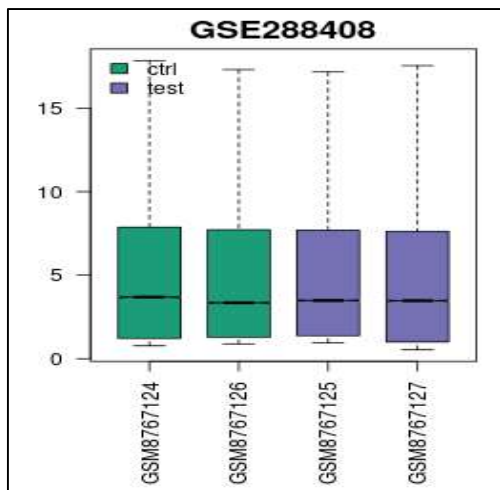


FIGURE 3.1: BOX PLOT

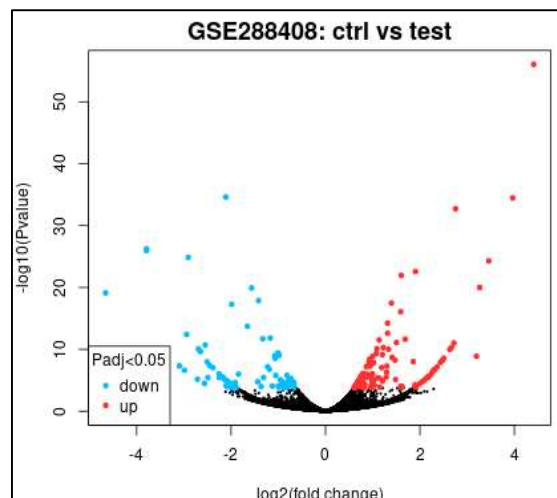


Figure 3.2 COMPARATIVE MEAN
DIFFERENCE PLOT

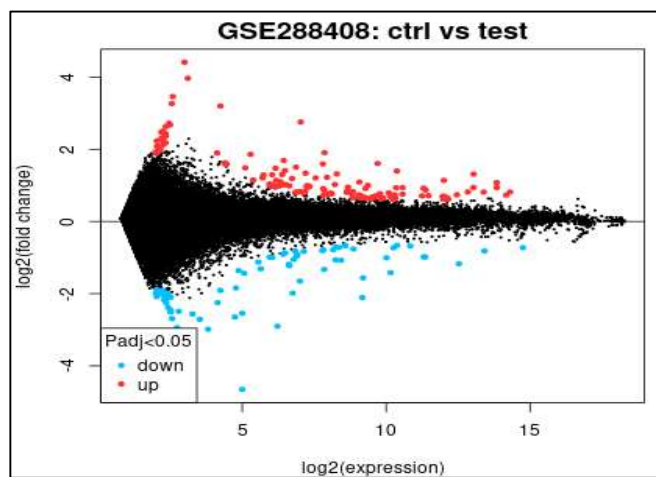


FIGURE 3.3 THE COMPARATIVE VOLCANO PLOT

TOP 10 HUB GENES

After the STRING server interaction, the protein network data was reported to Cytoscape analyzed with the packages namely Cytohubba, String network, y-field, algorithm. The top 10 hub genes are:



FIGURE 3.4 NETWORK AND PATHWAY ANALYSIS OF TOP 10 HUB GENES OBTAINED AND FOUND USING CYTOSCAPE.

PROTPARAM ANALYSIS

Result of ProtParam analysis of PPP2CB, in these various Physiochemical properties like the number of amino acid, molecular weight, theoretical pI, instability index, aliphatic index, hydropathicity of the compound are computed.

Number of amino acids: 309

Molecular weight: 35575.11

Theoretical pI: 5.21

Instability index:

The instability index (II) is computed to be 32.92

This classifies the protein as stable.

Aliphatic index: 78.19

Grand average of hydropathicity (GRAVY): -0.446

Figure 3.5 ProtParam analysis result of PPP2CBB

SOPMA RESULT

SOPMA was performed to determine the secondary structure of the receptor (hub genes):-

Alpha helix (Hh): 120 is 38.83%
 310 helix (Gg): 0 is 0.00%
 Pi helix (Ii): 0 is 0.00%
 Beta bridge (Bb): 0 is 0.00%
 Extended strand (Ee): 47 is 15.21%
 Beta turn (Tt): 0 is 0.00%
 Bend region (Ss): 0 is 0.00%
 Random coil (Cc): 142 is 45.95%
 Ambiguous states (?): 0 is 0.00%
 Other states : 0 is 0.00%

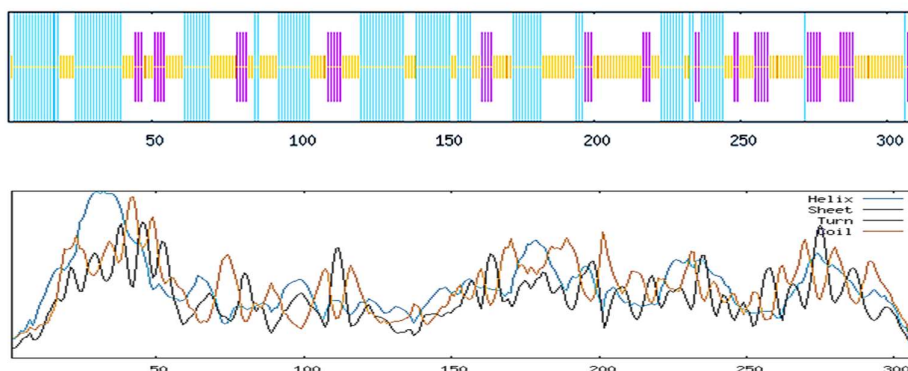


Figure 3.6: SECONADARY STRUCTURE OF PPP2CB

SOSUI RESULT

PPP2CB is SOLUBLE protein

Hydropathy & Charge plot

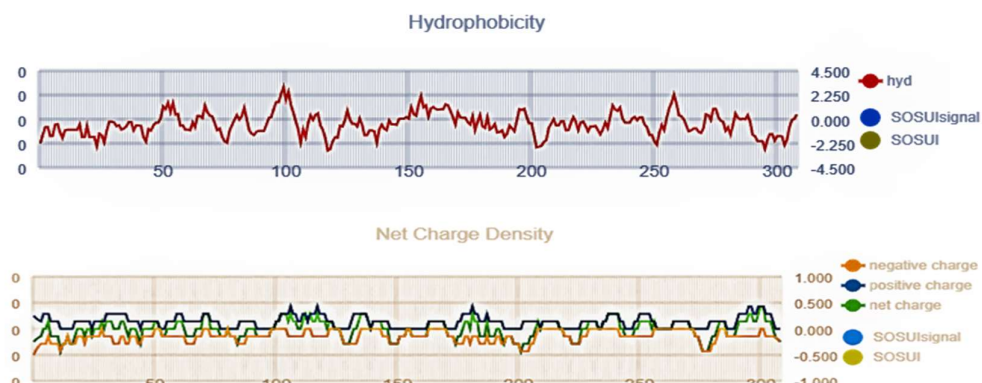


Figure 3.7: SOSUI RESULT OF PPP2CB

RAMACHANDRAN PLOT:

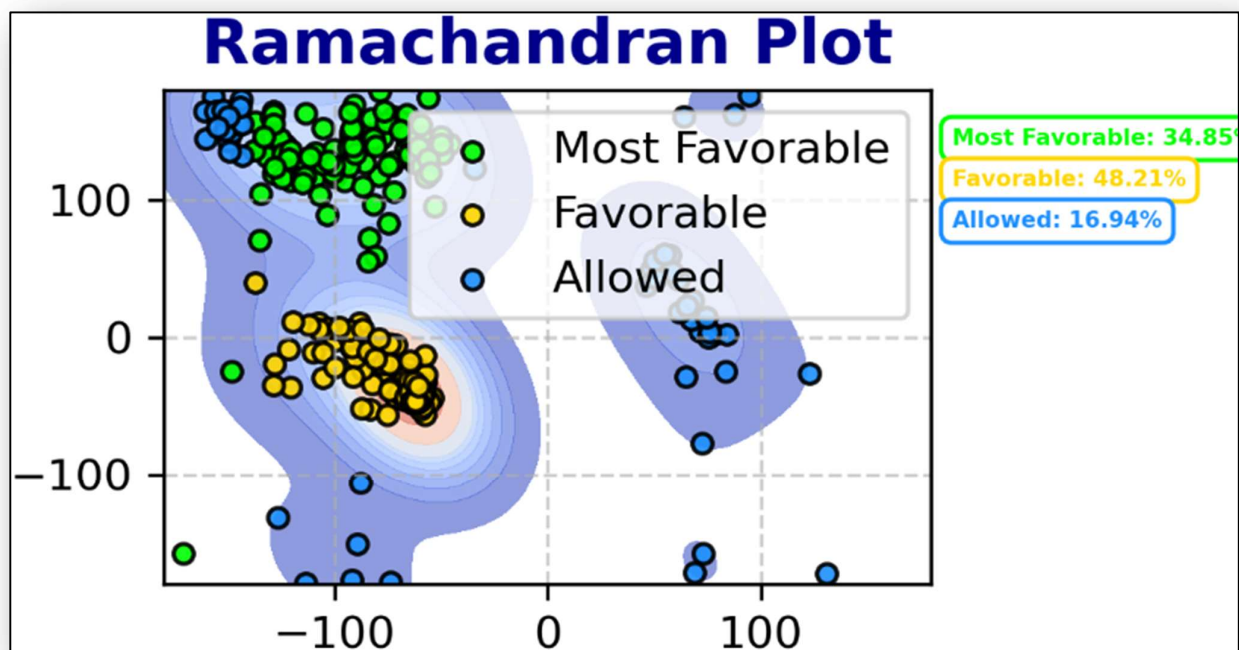


Figure 3.8: RAMACHANDRAN PLOT

TABLE 3.1: RESULT DRULITO ANALYSIS

SN o.	Title	M W	log p	Alo gp	H BA	H BD	TP SA	A M R	nR B	nAt om	nAci dic	R C	nRigi dB	nAr om	n H B	SAle rts
1	5281 614	286. 05	1.9 15	- 0.6 81	6	4	107. 22	81. 83	1	31	0	3	22	2	10	2
2	5280 445	286. 05	1.4 86	- 0.7 87	6	4	107. 22	81. 76	1	31	0	3	22	2	10	2

TABLE 3.2: OSIRIS TABLE

S. N o.	CI D	cLogP	SOLU BILTY	MOL WT	TPS A	DRUG LIKENES S	DRUG SCORE	MUTAGENIC	TUMORIGEN IC	IRRITA NT	REPRODUT IVE EFFECTIVE
1	5281614	1.84	-2.79	286.0	107.2	-0.07	0.4				
2	5280445	1.99	-2.56	26.0	107.2	1.9	0.84				

TABLE 3.3: DOCKING SCORE OF PHYTOCHEMICALS AS LIGANDS WITH PPP2CB AS PROTIEIN RECEPTOR

S. NO.	RECEPTOR + PHYTOCHEMICAL	DOCKING SCORE
1	PPP2CB + Fisetin	-5.630
2	PPP2CB + Luteolin	-5.145

TABLE 4: DOCKING RESULTS OF HUB GENES OF Fisetin

Name	Visible	Color	Parent	Distance	Category	Types	From	From Chemistry	To	To Chemistry	Angle DHA	Angle HAY	Theta
1 :UNL1:H - A:LEU183:O	☒ Yes		Ligand Non-bond Monitor	1.9646	Hydrogen Bond	Conventional Hydrogen Bond	:UNL1:H	H-Donor	A:LEU183:O	H-Acceptor	132.61	130.578	
2 :UNL1:H - A:ASP184:OD1	☒ Yes		Ligand Non-bond Monitor	2.56683	Hydrogen Bond	Conventional Hydrogen Bond	:UNL1:H	H-Donor	A:ASP184:OD1	H-Acceptor	104.59	114.383	
3 :UNL1:H - A:LYS41:O	☒ Yes		Ligand Non-bond Monitor	1.96833	Hydrogen Bond	Conventional Hydrogen Bond	:UNL1:H	H-Donor	A:LYS41:O	H-Acceptor	128.253	149.14	
4 A:ARG185:NH1 - :UNL1	☒ Yes		Ligand Non-bond Monitor	4.31074	Electrostatic	Pi-Cation	A:ARG185:NH1	Positive	:UNL1	Pi-Orbitals			36.656

Table 3.5 DOCKING RESULTS OF HUB GENES OF LUTEOLIN

Name	Visible	Color	Parent	Distance	Category	Types	From	From Chemistry	To	To Chemistry	Angle DHA	Angle HAY	Theta
1 A:ASN44:HD21 - :UNL1:O	☒ Yes		Ligand Non-bond Monitor	2.84629	Hydrogen Bond	Conventional Hydrogen Bond	A:ASN44:HD21	H-Donor	:UNL1:O	H-Acceptor	133.552	110.544	
2 :UNL1:H - A:ASP184:OD1	☒ Yes		Ligand Non-bond Monitor	2.53069	Hydrogen Bond	Conventional Hydrogen Bond	:UNL1:H	H-Donor	A:ASP184:OD1	H-Acceptor	129.995	119.181	
3 :UNL1:H - A:LYS41:O	☒ Yes		Ligand Non-bond Monitor	2.74507	Hydrogen Bond	Conventional Hydrogen Bond	:UNL1:H	H-Donor	A:LYS41:O	H-Acceptor	100.494	151.029	
4 A:ASP184:HA - :UNL1:O	☒ Yes		Ligand Non-bond Monitor	2.72187	Hydrogen Bond	Carbon Hydrogen Bond	A:ASP184:HA	H-Donor	:UNL1:O	H-Acceptor	138.425	105.048	
5 A:ARG185:NH1 - :UNL1	☒ Yes		Ligand Non-bond Monitor	4.198	Electrostatic	Pi-Cation	A:ARG185:NH1	Positive	:UNL1	Pi-Orbitals			38.868

VISUALIZATION OF 2-D AND 3-D STRUCTURE OF DOCKING

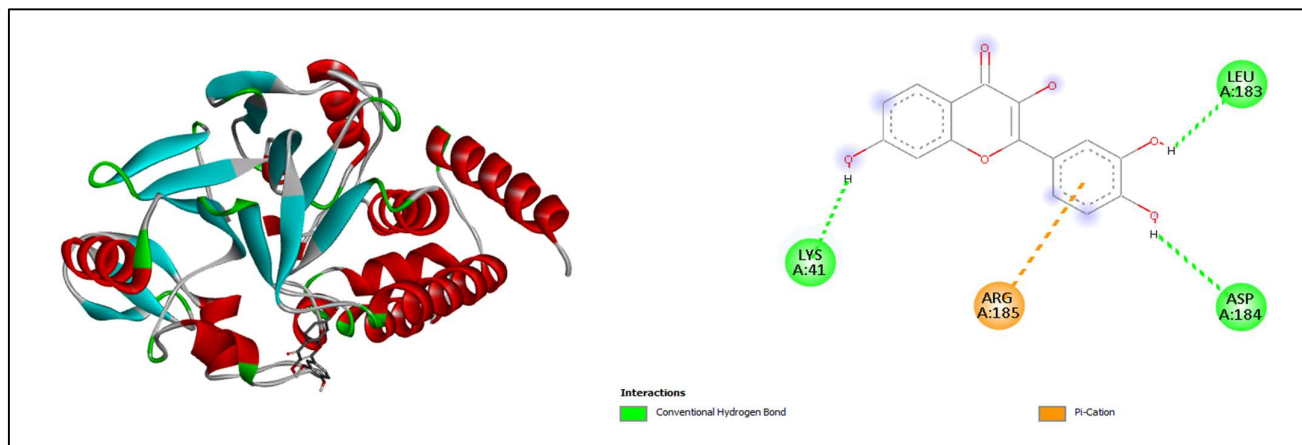


Figure 3.9 PPP2CB AND FISETIN

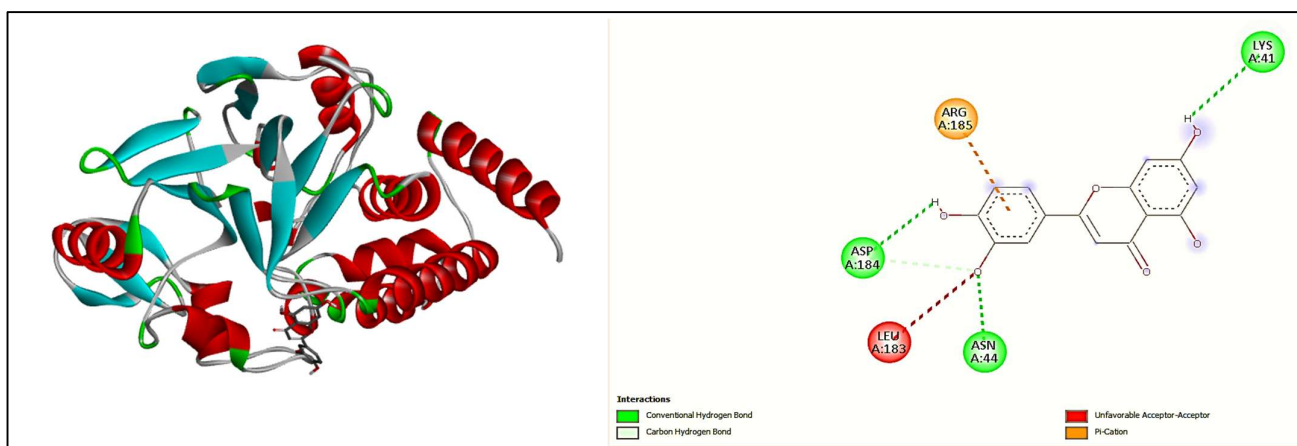


Figure 3.10: PPP2CB AND LUTEOLIN

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